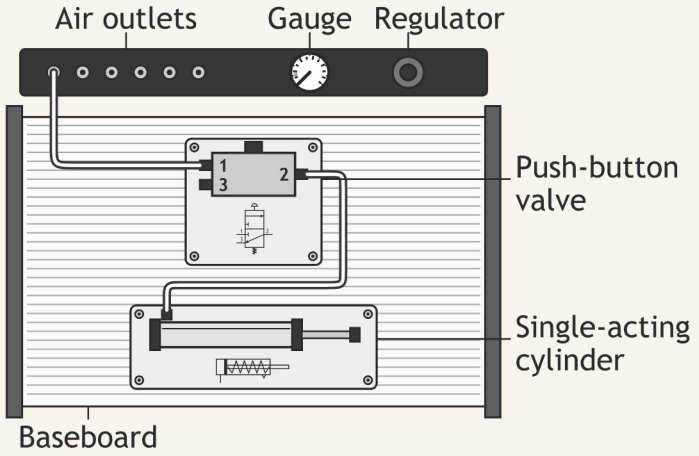
Daily Review



1. Name one thing air can move.
2. Three parts of a system?
3. Which engineer works on moving parts?
4. Energy of movement is called?
5. Energy is measured in?

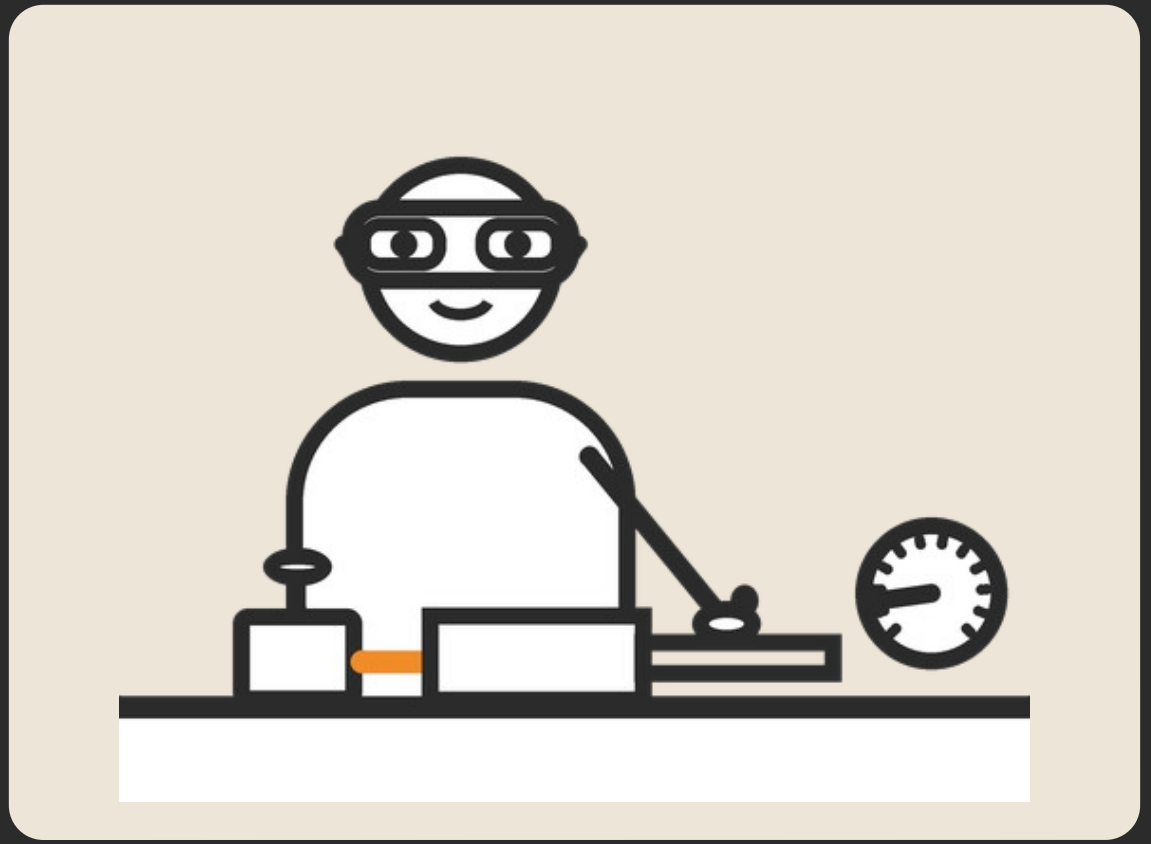
Safety First and Last

How do you make a park gate move with air, safely?



S3 Engineering Science

Squashed air can push hard. How do we use it safely?



The danger: a hand on the rod

PART 1

Safety First and Last

Learn to use air safely, then build your first air circuit.

By the end of this part…



1. I can follow the safety rules and the regulator routine
2. I can say what the rod does when I press and let go
3. I can test a circuit in the simulation before I build it

SECTION A

Safety First

Air can push with a big force. What keeps us safe?

Air Can Push Hard

Pneumatics

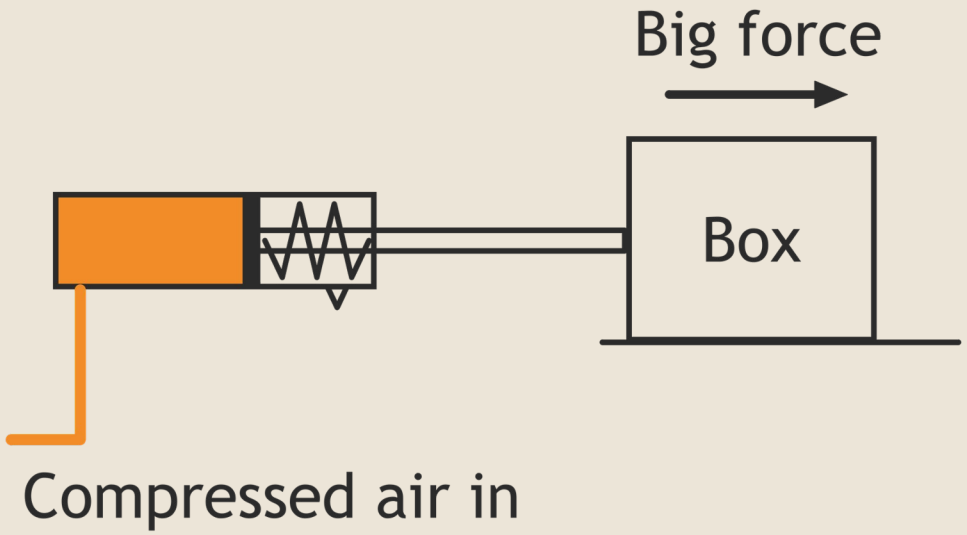
using compressed air to make things move

Compressed air

air squashed into a small space

Why take care

it pushes with a big force. It can hurt eyes, ears and skin.



Where You Meet It: Air Tools

Air tools

garages undo wheel nuts with air

Why air

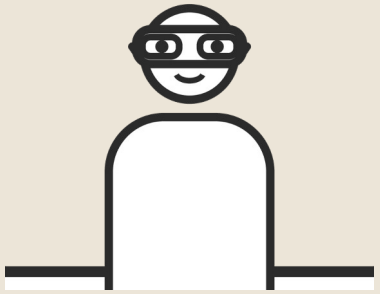
it pushes hard and fast

Take care

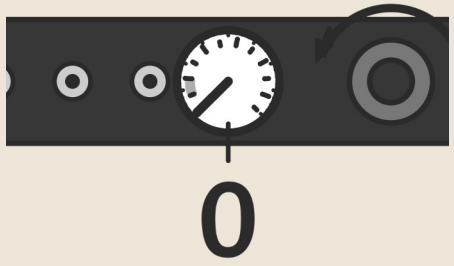
the same five safety rules



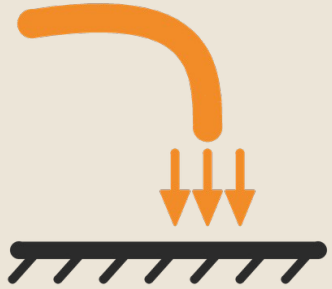
The 5 Safety Rules (1 of 2)



1 Goggles on when you build and when you test. A tube can blow off near your eyes.



2 Regulator at 0 while you build or change a circuit. Parts can move while your hands are in.

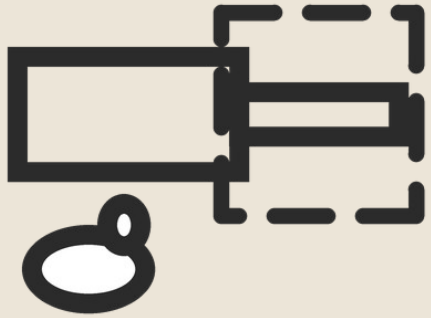


3 Never point air at anyone. Tube ends point down at the bench.

The 5 Safety Rules (2 of 2)



4 Hear a hiss? Regulator to 0 at once. A hiss is a leak. The tube may blow off.



5 Fingers clear of the piston rod. It moves out with force. It can trap fingers.

The Gauge: Between 1 and 2 Bar

Read

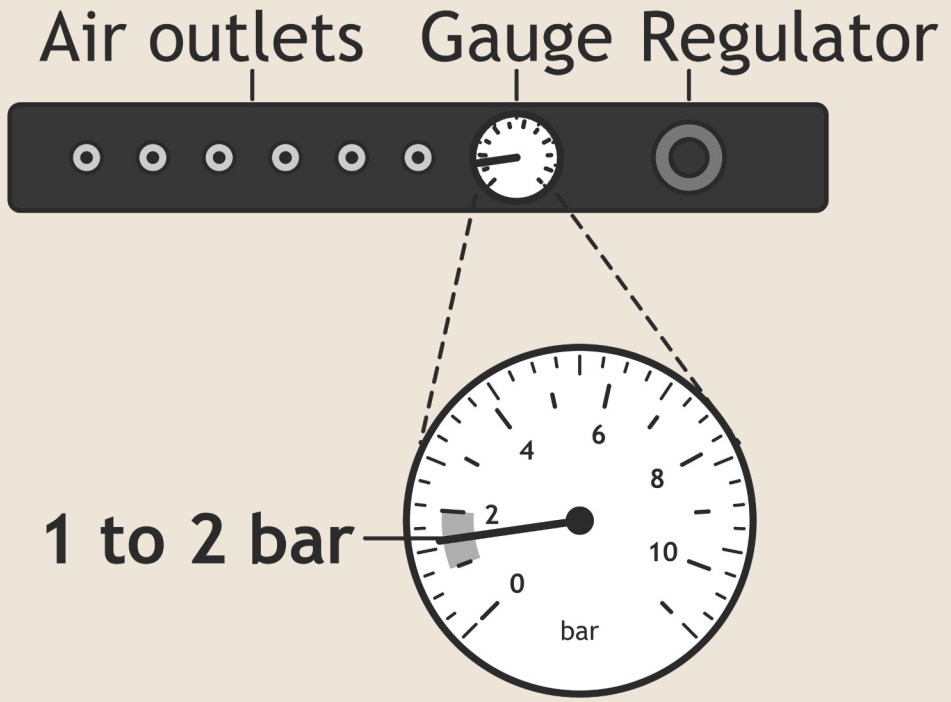
the inner numbers. They are in bar.

Set

between 1 and 2 bar. Never above 2. Never below 1.

Change anything?

turn it back to 0 first.



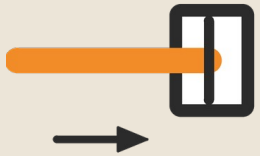
The Regulator Routine



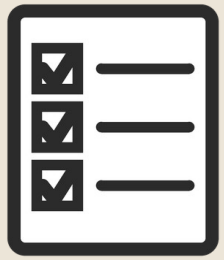
1 Goggles on



2 Regulator at 0



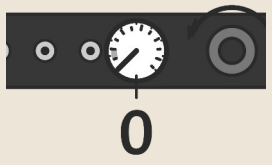
3 Build
push every tube in fully



4 Air Permit signed



5 Set 1 to 2 bar
then test



6 Back to 0
regulator to 0 when you finish

Show Me: Safety Quick-Fire



1. Before you change a tube, the regulator is at…
2. You hear a hiss. What do you do first?
3. Where do tube ends point?
4. What pressure do you set for testing?
5. When do you wear goggles?

You want to move a valve. The air is on. What do you do first?



A move it quickly

B turn the regulator to 0

C ask a friend to hold the cylinder

D press the button to empty it

Safety Licence sheet



On your own, in silence. 5 questions.
You need 4 out of 5 to pass.

3 min

SECTION B

Your Team and First Job

Four roles, one client, one job.

Team Roles

Rotate each task. In a group of 3, one pupil gets 2 tasks.



Safety
reads the routine, signs the Air Permit, says Stop



Builder
builds the circuit on the kit



Tester
builds and tests it in the simulation first



Recorder
reads out the results for Compare

Your Client: Harbourlight Adventure Park

Client

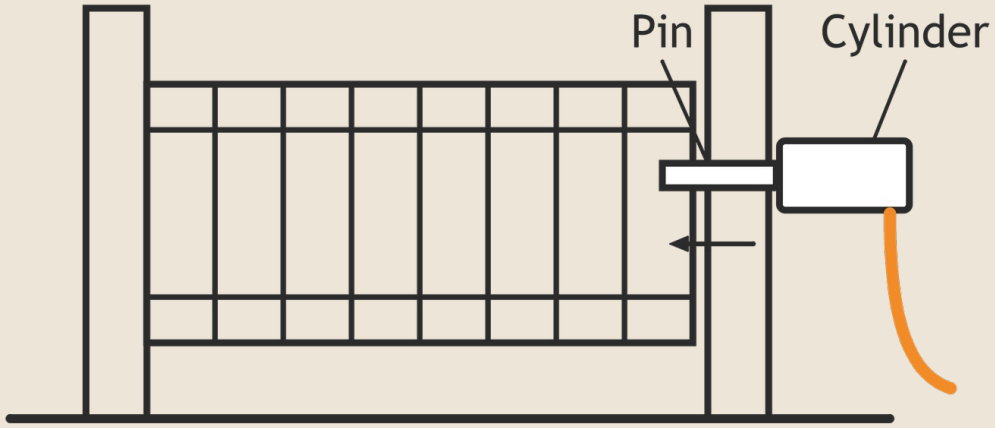
Jo Kerr, Harbourlight's chief engineer

Job 1

the ticket gate: a button unlocks it

Done when

the pin goes out, then back in



SECTION C

The Ticket Gate

Simulate. Air Permit. Then the kit.

The Gate Circuit

Regulator

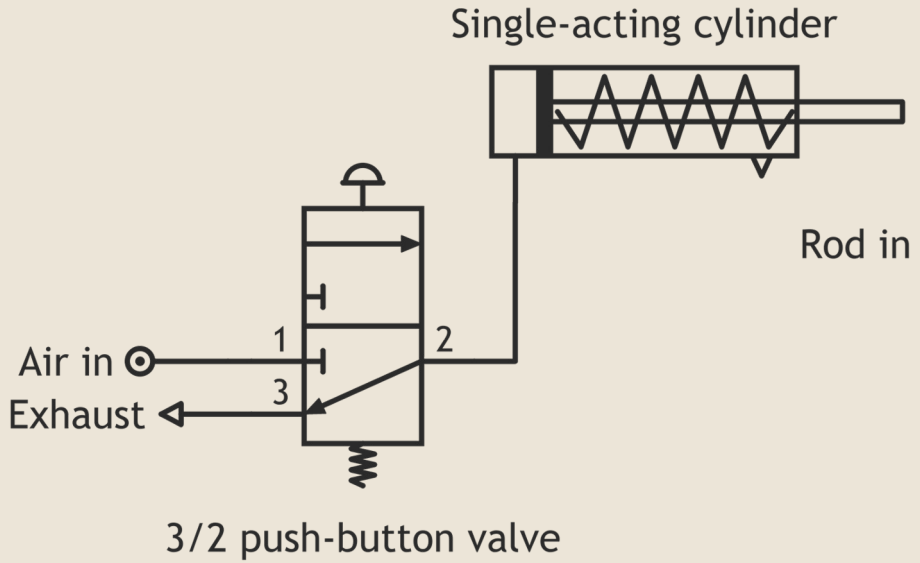
how hard the air pushes

3/2 push-button valve

input: 3 ports, 2 states

Single-acting cylinder

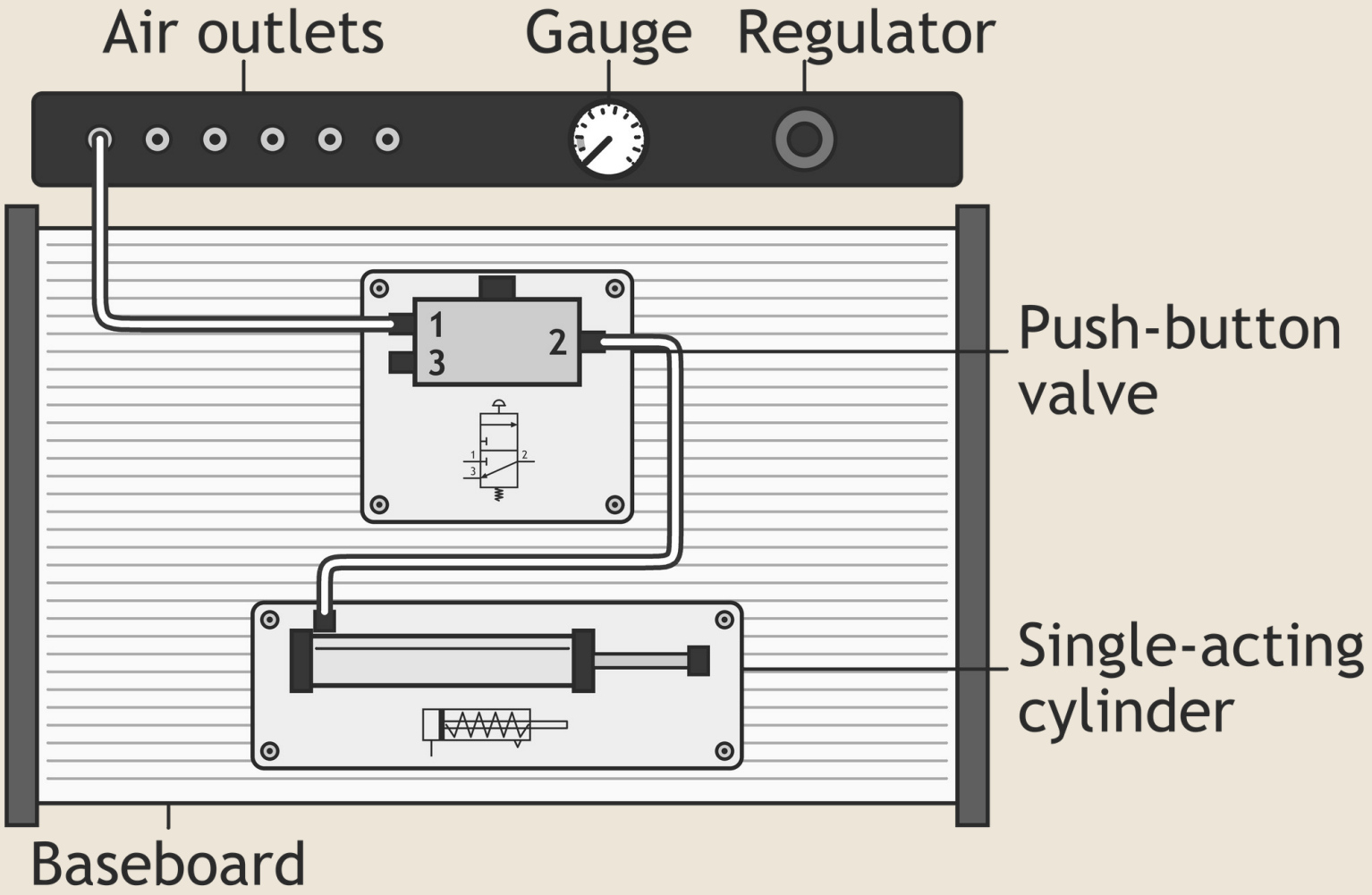
output: air pushes the rod out



3/2 push-button valve

Air supply: the circle with a dot, where the air comes in.

The Kit, Seen From Above



Predict

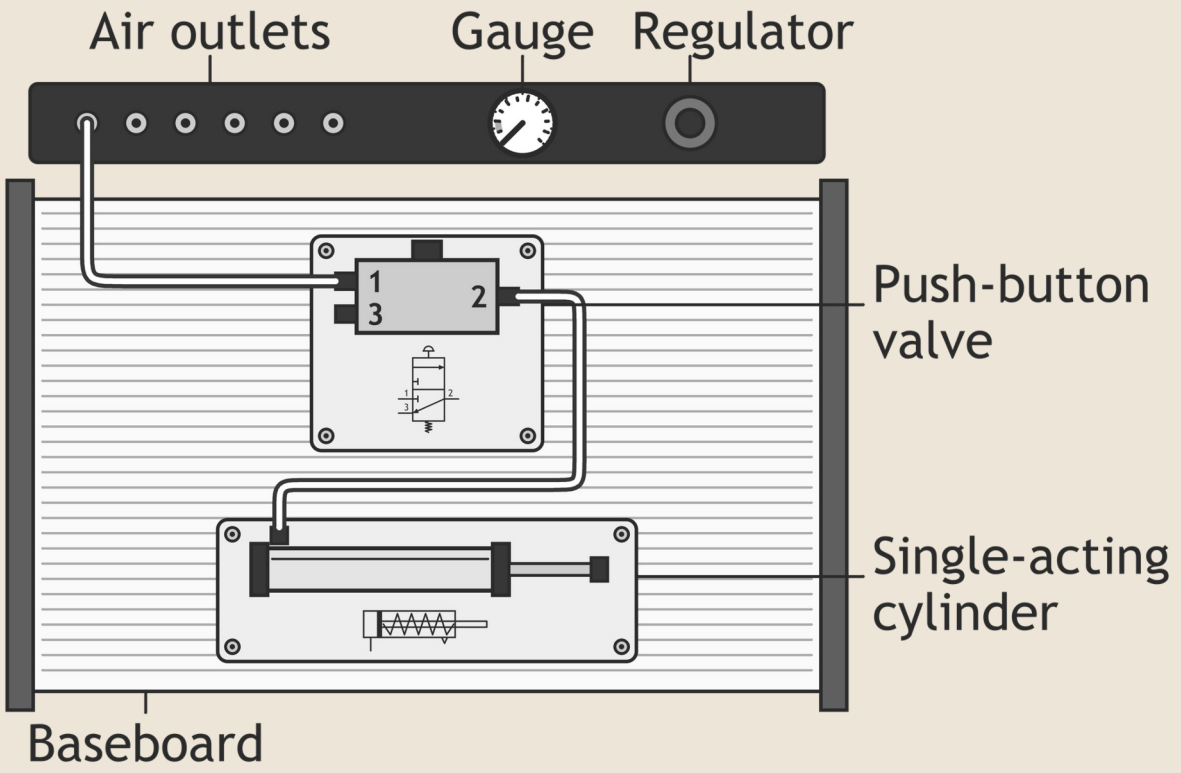
1. You press and hold the button. What does the rod do?

2. You let go of the button. What does the rod do?

What to Collect

Builder and Safety: one set per team.

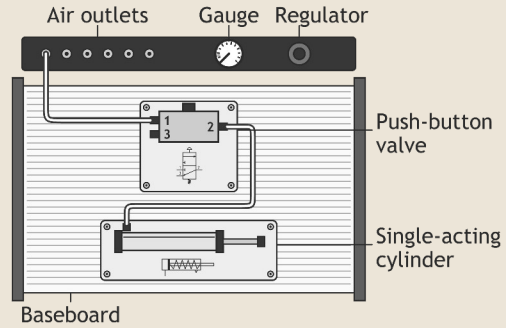
1 Bench card 1
2 Manifold (air supply)
3 3/2 push-button valve
4 Single-acting cylinder
5 Tubing
6 Goggles, one pair each



Simulate and build

Tasks 1 and 2, page 2

1 Tester and Recorder: run the simulation.
2 Builder and Safety: collect, then build.
3 Safety: check the safety list.
4 Teacher check, then the air goes on.

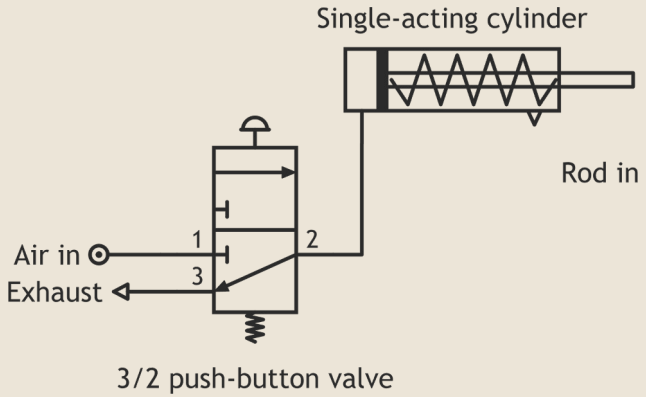


10 min

Compare and Questions

Tasks 3 to 5
booklet page 3

Task 3: fill in Compare.
Task 4: Questions 1 to 3 on your own.
Task 5: Key words.



5 min

Answers: Questions and Key Words

1. 1 Input and output of the gate circuit
2. 2 Press and hold, then let go
3. 3 Why regulator to 0 before a change?
4. Key words

Plenary: Safety First and Last

1. What pressure do we test at?
2. What must you do before you change a tube?
3. Which part is the input in the gate circuit?

Plenary: Safety First and Last

4 What pulls the rod back in?

5 What must a circuit do before it gets air?

Look back — by the end of this part…



1. I can follow the safety rules and the regulator routine
2. I can say what the rod does when I press and let go
3. I can test a circuit in the simulation before I build it

LOOKING AHEAD

Lesson 2: Reading the Diagram

Symbols, ports and states: what happens when you let go?

Credits

The 5 Safety Rules: Safety rules and the Lesson 1 circuit adapted from the Matrix CP2080 Pneumatics Essentials worksheets (© Matrix Technology Solutions Limited). Drawings by R Stewart.

R Stewart - October 2026